

Moore's Law and Its Physical Limits

Moore's Law is the 1965 observation by Intel co-founder Gordon Moore that the number of transistors packed onto a microchip doubles roughly every two years, exponentially increasing computing power while cutting costs. It is an economic and engineering roadmap rather than a literal law of physics.

Today, Moore's Law has effectively hit a wall due to several distinct physical limitations:

- **Atomic and Quantum Scale Limits:** Transistors have shrunk down to the scale of just a few nanometers—roughly the width of a few silicon atoms. At this size, electrons begin to misbehave due to **quantum tunneling**. They breach the microscopic insulating barriers even when the transistor is turned "off," creating massive electrical leakage and rendering the switch unreliable.
- **The Power Wall:** As transistor density increased, engineering teams hit the limits of **Dennard Scaling**. Historically, shrinking transistors allowed for a proportional reduction in voltage. However, voltage cannot drop below the threshold required to reliably switch a transistor without succumbing to background electrical noise.
- **Thermal Dissipation Limits:** Because voltage can no longer be lowered, packing billions of tightly jammed transistors into the same small footprint causes **dynamic power consumption** and heat generation to skyrocket. This creates unsustainable thermal density, meaning chips would literally melt if driven at higher clock speeds, freezing consumer CPU frequencies around 4 to 5 GHz.
- **Economic Limitations:** The infrastructure needed to manufacture sub-3nm nodes requires extreme ultraviolet (EUV) lithography machines costing hundreds of millions of dollars. The skyrocketing cost of fabrication makes continued exponential shrinking economically unviable for most industries.

Instead of shrinking single transistors further, the tech industry has shifted toward multi-core architectures, specialized accelerators (like GPUs), and 3D chip stacking to advance computing power.